

APM1110

Probability & Probability Distribution

FA7

An Analysis on FEU Manila Students' Allotted Lunch Money

Diche, Jasmine

Hermosada, Joshua

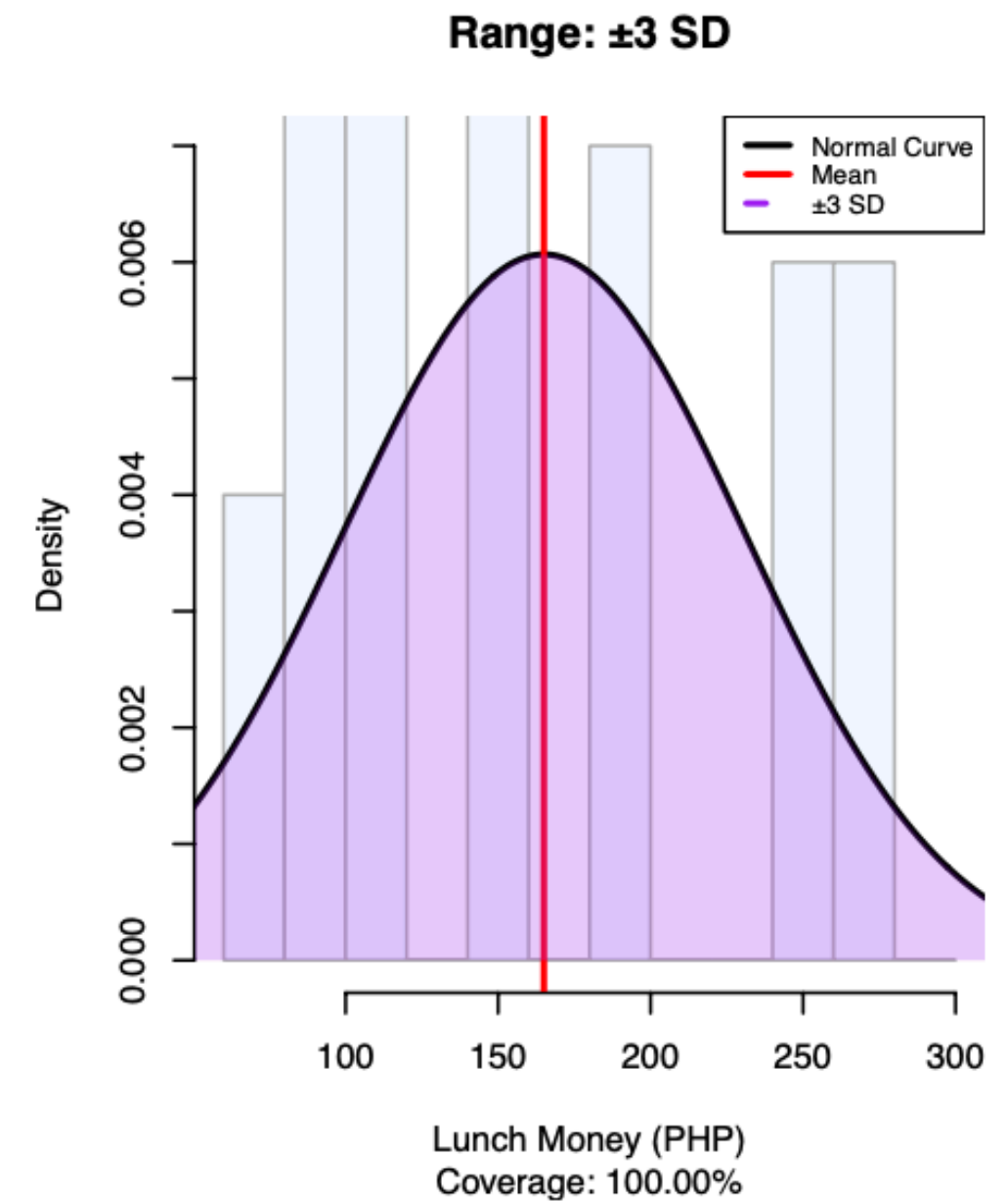
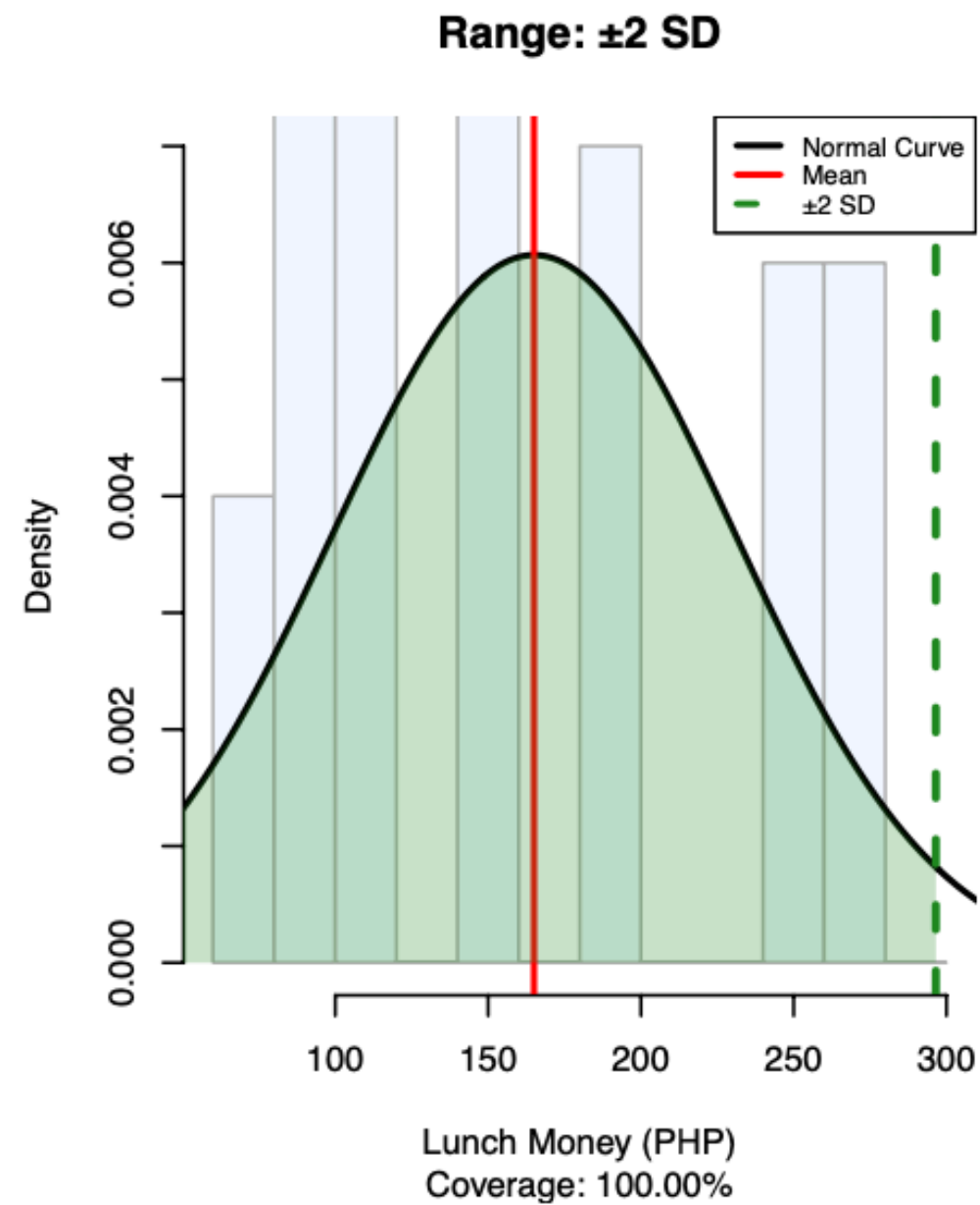
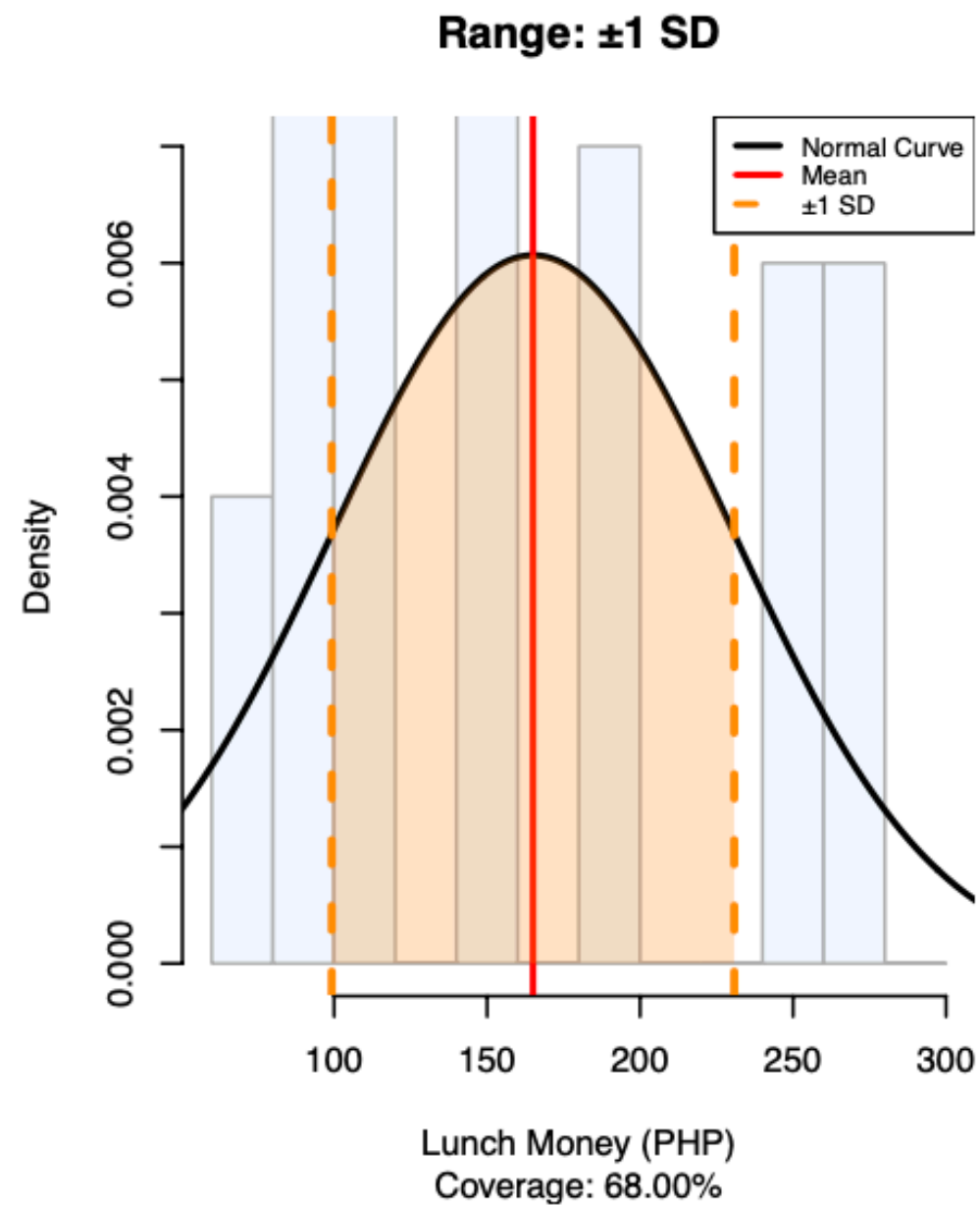
Methodology

- To ensure a meaningful sample size, we surveyed **exactly 50 students** across the campus
- We collected this data verbally to keep the process quick and non-intrusive. To respect privacy and follow ethical guidelines, we did not record any names. Instead, we digitized our findings inside Excel using simple student IDs from 1 to 50 before running our analysis in R.

Statistics of Lunch Allowances

- Mean: 165.00 PHP
- Population Standard Deviation: 65.76 PHP
- Sample Size: 50 students

R Plots



Interpretation

- Is the distribution symmetric?

The distribution is relatively symmetric and tightly balanced around the center, but it features a flatter, uniform top

22 students fall below the mean (spending 80, 100, or 120 PHP), 19 students sit above it (spending 200, 250, or 280 PHP), and 9 are right in the middle at 150 PHP. This balanced weight on both sides allows it to fit a normal curve beautifully.'

- Are there outliers?

The data contains absolutely zero outliers. Because 100% of our students fall safely within 2 standard deviations (the 33.47 to 296.53 PHP window), there are no extreme, isolated anomalies pulling or distorting our graph.

Interpretation

- What does the shape of the distribution imply?

The flat-topped, widespread shape indicates that student purchasing power is evenly distributed across a wide financial spectrum, rather than everyone crowding around a single price tag."

- How can this data be useful?

Since 68% of the student body works with a budget between 99.24 PHP and 230.76 PHP, vendors can capture the largest chunk of the market by keeping their standard meal combinations priced inside this exact sweet spot—ideally around 120 to 150 PHP.

Our frequency data shows that 24% of the students surveyed survive on a lunch budget of 100 PHP or less. This highlights an opportunity for the student welfare committee to ensure the canteen continues providing nutritious, budget-friendly choices to support lower-income brackets.

Thank You!